

6. Project Development Phase:

Project Description:

The Farm Management System (FMS) in Salesforce is a comprehensive solution designed to digitize and streamline agricultural operations. This application empowers farmers, farm managers, and stakeholders by managing data related to crops, Farm, Farmer, and buyers, while providing actionable insights through reports, and dashboards.

Milestone 1-Salesforce:

What Is Salesforce?

Salesforce is your customer success platform, designed to help you sell, service, market, analyze, and connect with your customers.

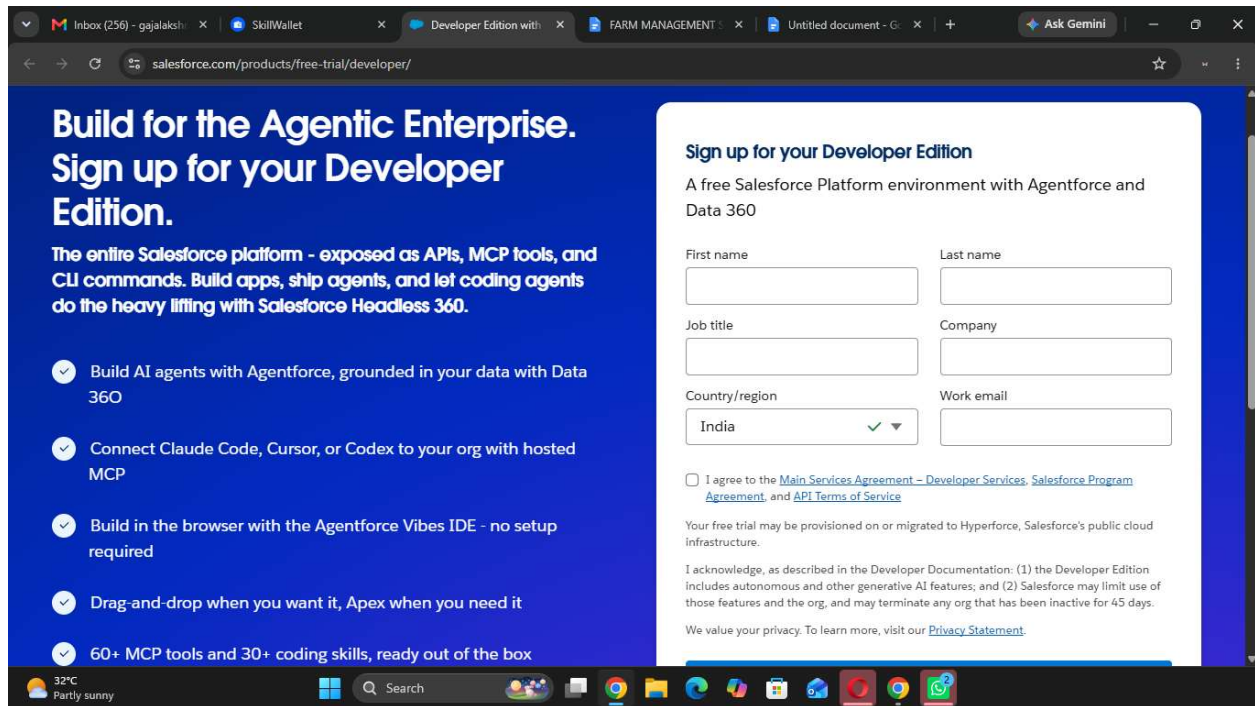
Salesforce has everything you need to run your business from anywhere. Using standard products and features, you can manage relationships with prospects and customers, collaborate and engage with employees and partners, and store your data securely in the cloud.

Activity1:

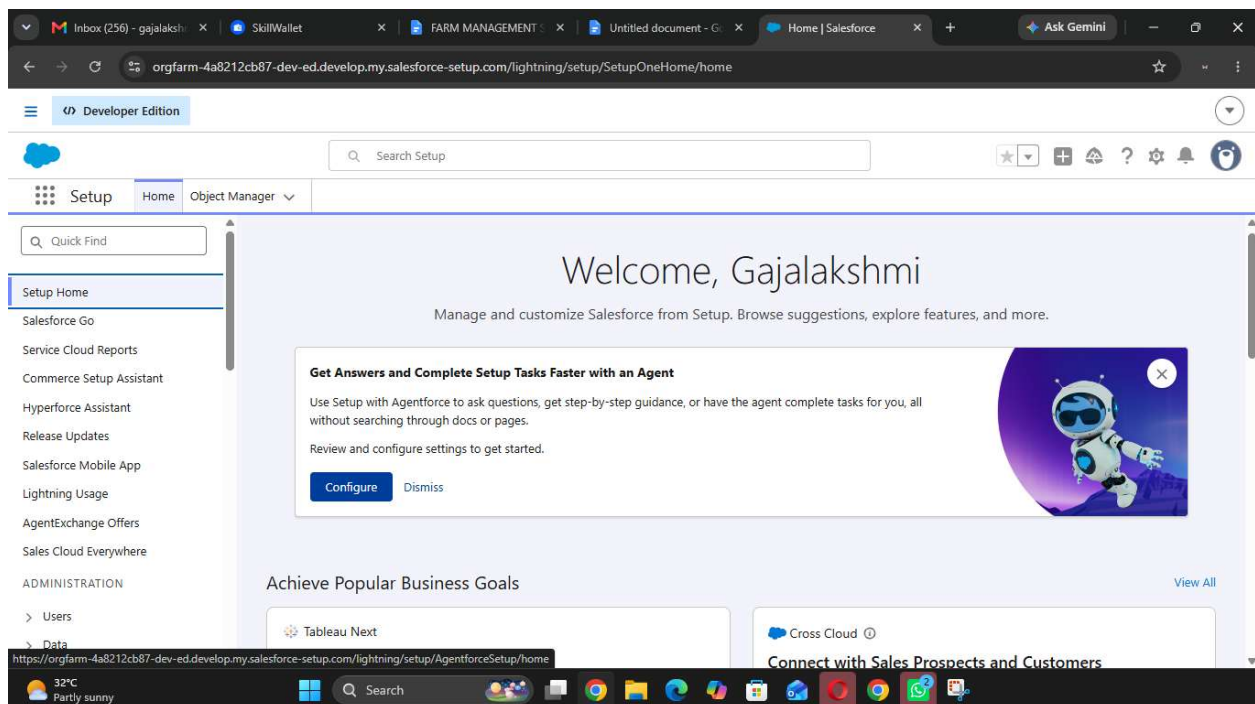
Creating a Salesforce Developer Org:

A Developer org has all the features and licenses you need to get started with Salesforce.

1. Search [Developer.salesforce.com/signup](https://developer.salesforce.com/signup)
2. Enter the following details like First name, last name, Email, Role, Company, Country/Region, Postal code, and Username must be unique.
3. Click sign me up, after a few min you will receive a mail from salesforce org and by using the verify account link you can create your new password.
4. Click save.
5. Search login.salesforce.com



6. By using username and password you can into the Salesforce org. The homepage will appear below



Milestone 2-Object:

What is an object?

Salesforce objects are database tables that permit you to store data that is specific to an organization. It consists of fields (columns) and records (rows).

In This Application We Use 4 Custom Objects:

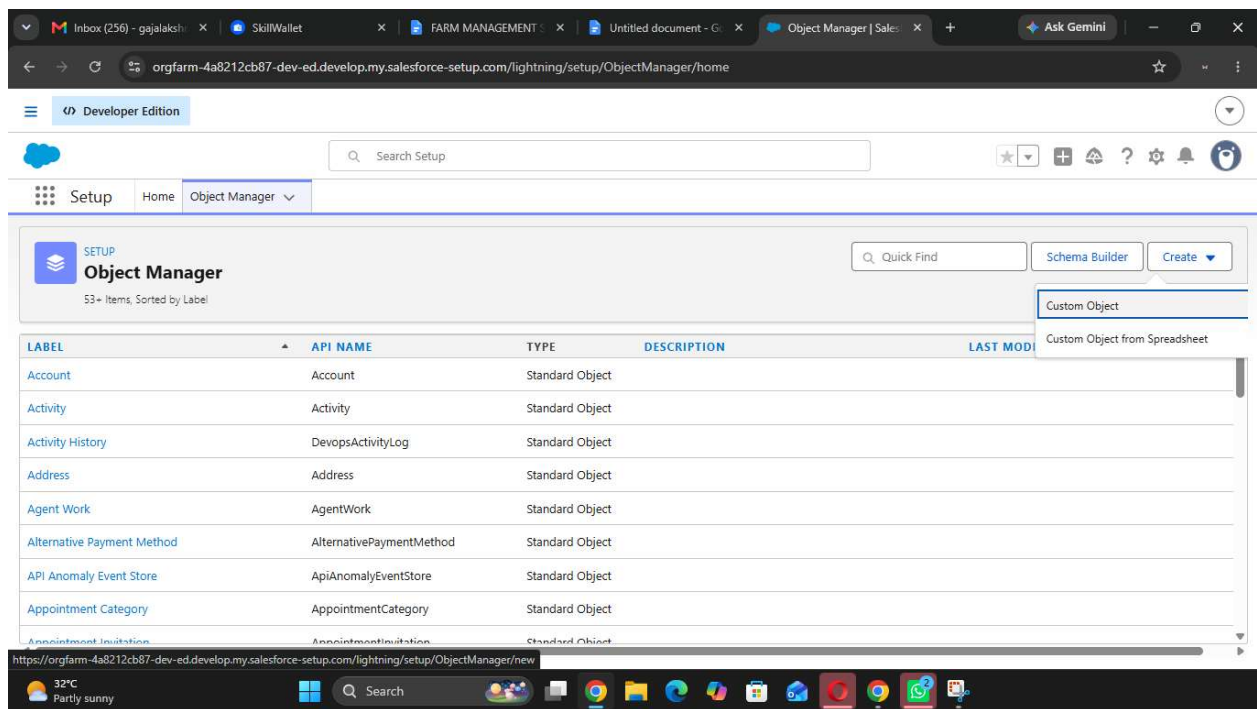
- 1.Farm
- 2.Crop
- 3.Farmer
- 4.Buyer

Activity 1:

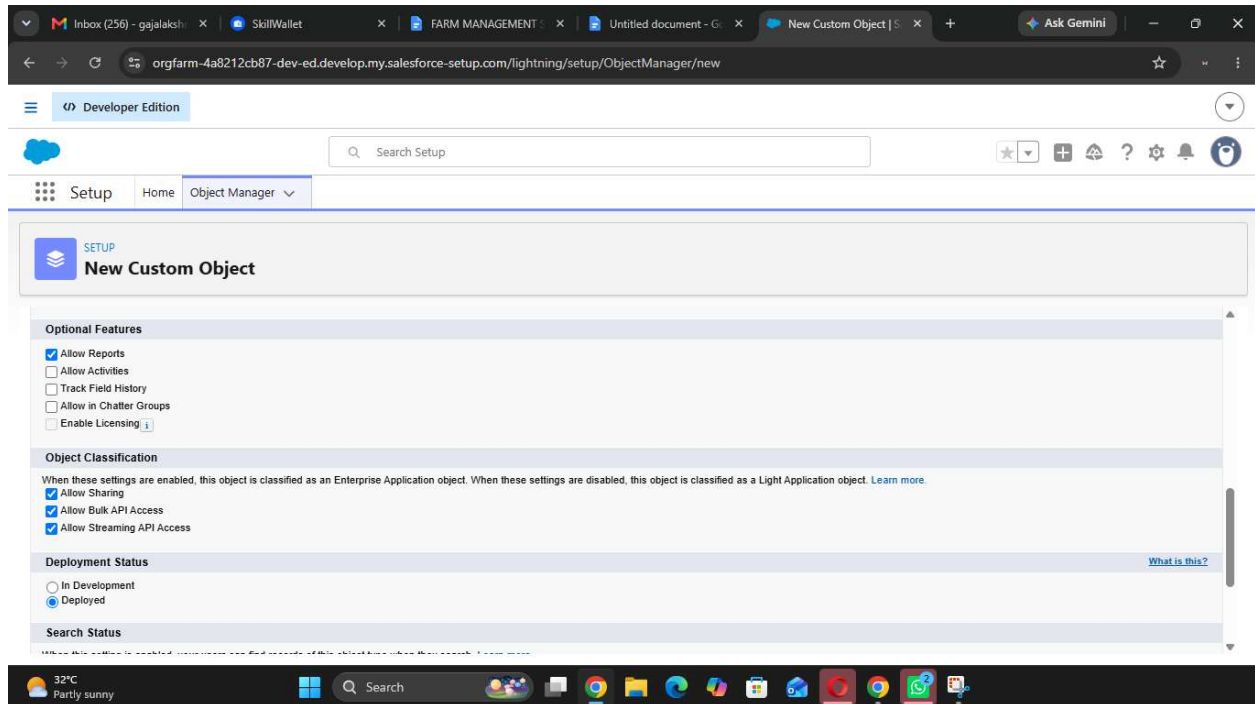
Create a custom object for Farm:

To create a custom object, follow these steps:

1. From setup click on object manager.
2. Click create, select custom object.



3. Fill in the label as " Farm ".
4. Fill in the plural label as " Farms".
5. Record name: Farm Name"
6. Select the data type as "Text".



7. In the Optional Features section, select Allow Reports and Track Field History.
8. In the Deployment Status section, ensure Deployed is selected.
9. In the Search Status section, select Allow Search.
10. In the Object Creation Options section, select Add Notes and Attachments related list to default page layout.
11. Leave everything else as is, and click Save.

Activity 2:

Creation of Crop object:

1. Click on the gear icon and then select Setup.
2. Click on the object manager tab just beside the home tab.
3. After the above steps, have a look on the extreme right you will find a Create Dropdown click on that and select Custom Object.
 - a. On the Custom Object Definition page, create the object as follows:
 - b. Label: Crop

- c. Plural Label: Crops
- d. Record Name: Crop Name
- e. Select the data type as "Text".
- f. Check the Allow Reports checkbox g. Check the Allow Search checkbox
- h. In the Object Creation Options section, select Add Notes and Attachments related list to default page layout i. Click Save.

Activity 3:

Creation of Farmer object:

- 1. Click on the gear icon and then select Setup.
- 2. Click on the object manager tab just beside the home tab.
- 3. After the above steps, have a look on the extreme right you will find a Create Dropdown click on that and select Custom Object.
- a. On the Custom Object Definition page, create the object as follows:
- b. Label: Farmer
- c. Plural Label: Farmers
- d. Record Name: Farmer Name e. Select the data type as "Text".
- f. Check the Allow Reports checkbox. g. Check the Allow Search checkbox.
- h. In the Object Creation Options section, select Add Notes and Attachments related list to default page layout. i. Click Save.

Activity 4:

Creation of Buyer object:

- 1. Click on the gear icon and then select Setup.
- 2. Click on the object manager tab just beside the home tab.
- 3. After the above steps, have a look on the extreme right you will find a Create Dropdown click on that and select Custom Object.
- a. On the Custom Object Definition page, create the object as follows:
- b. Label: Buyer
- c. Plural Label: Buyers
- d. Record Name: Buyer Name
- e. Select the data type as "Text".
- f. Check the Allow Reports checkbox. g. Check the Allow Search checkbox.

- h. In the Object Creation Options section, select Add Notes and Attachments related list to default page layout.
- i. Click Save.

Milestone 3 - Tab: Tab

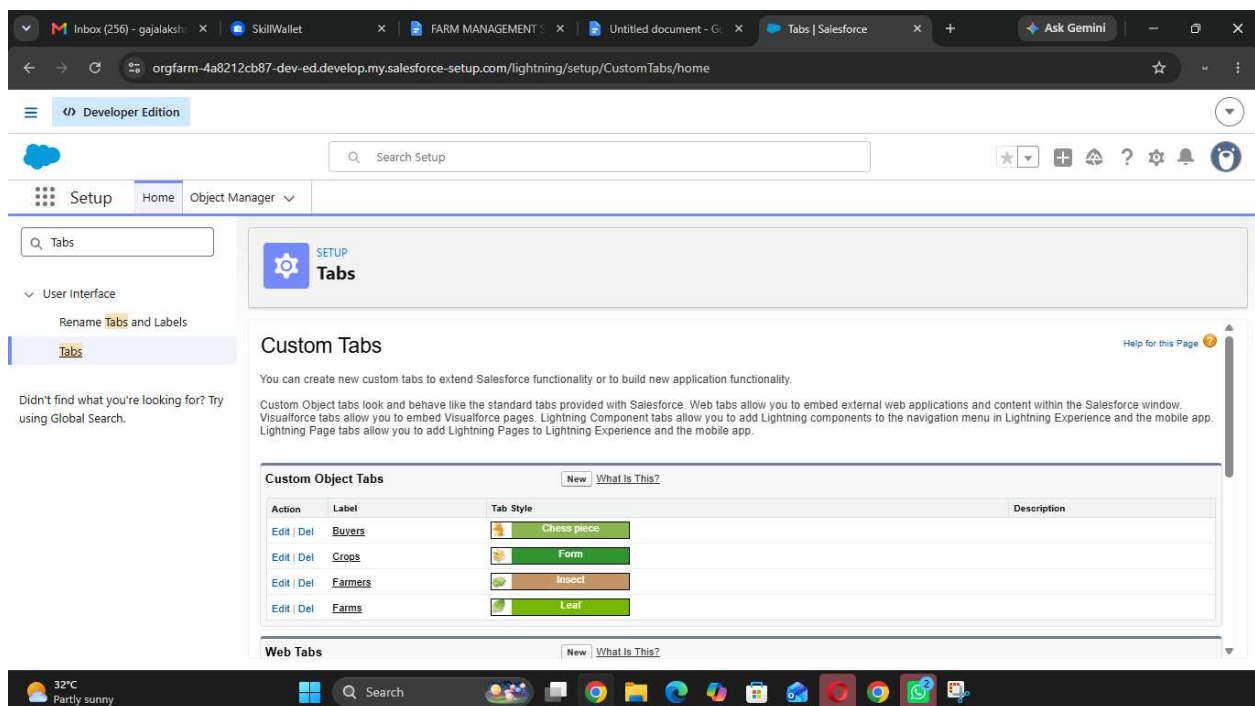
Tabs in Salesforce help users view the information at a glance. It displays the data of objects and other web content in the application.

Activity 1:

Creation of Farm tab:

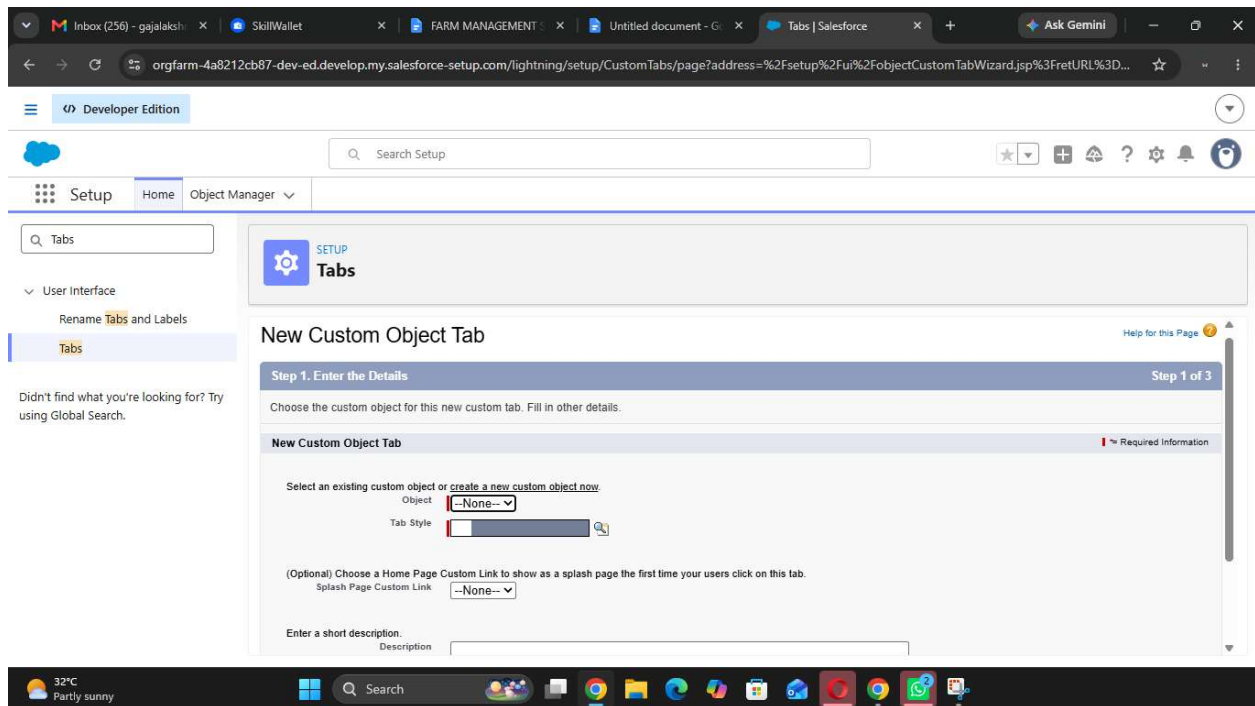
Now create a custom tab.

1. Click on the Home tab, enter Tabs in Quick Find and select Tabs.



2. Under custom object tabs, click New.

3. For Object, select Farm.
4. For Tab Style, select any icon.
5. Leave all defaults as is. Click Next, Next, and Save.



Activity 2:

Creation of Crop tab:

Now create a custom tab.

1. Click on Home tab, enter Tabs in Quick Find and select Tabs.
2. Under custom object tabs, click New.
3. For Object, select Crop.
4. For Tab Style, select any icon.
5. Leave all defaults as is. Click Next, Next, and Save.

Activity 3:

Creation of Farmer tab:

Now create a custom tab.

1. Click on the Home tab, enter Tabs in Quick Find and select Tabs.
2. Under custom object tabs, click New.
3. For Object, select Farmer.
4. For Tab Style, select any icon.
5. Leave all defaults as is. Click Next, Next, and Save.

Activity 4:

Creation of Buyer tab:

Now create a custom tab.

1. Click on Home tab, enter Tabs in Quick Find and select Tabs.
2. Under custom object tabs, click New.
3. For Object, select Buyer
4. For Tab Style, select any icon.
5. Leave all defaults as is. Click Next, Next, and Save.

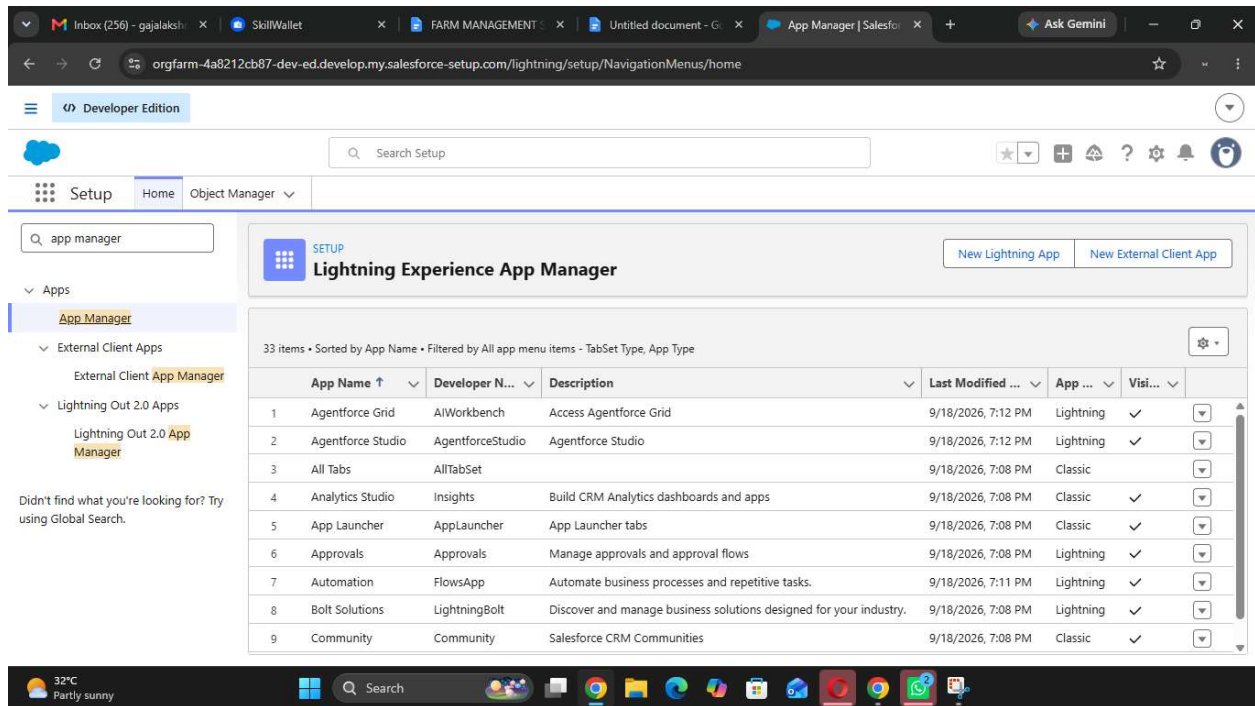
Milestone 4 - Lightning App:

What is an App?

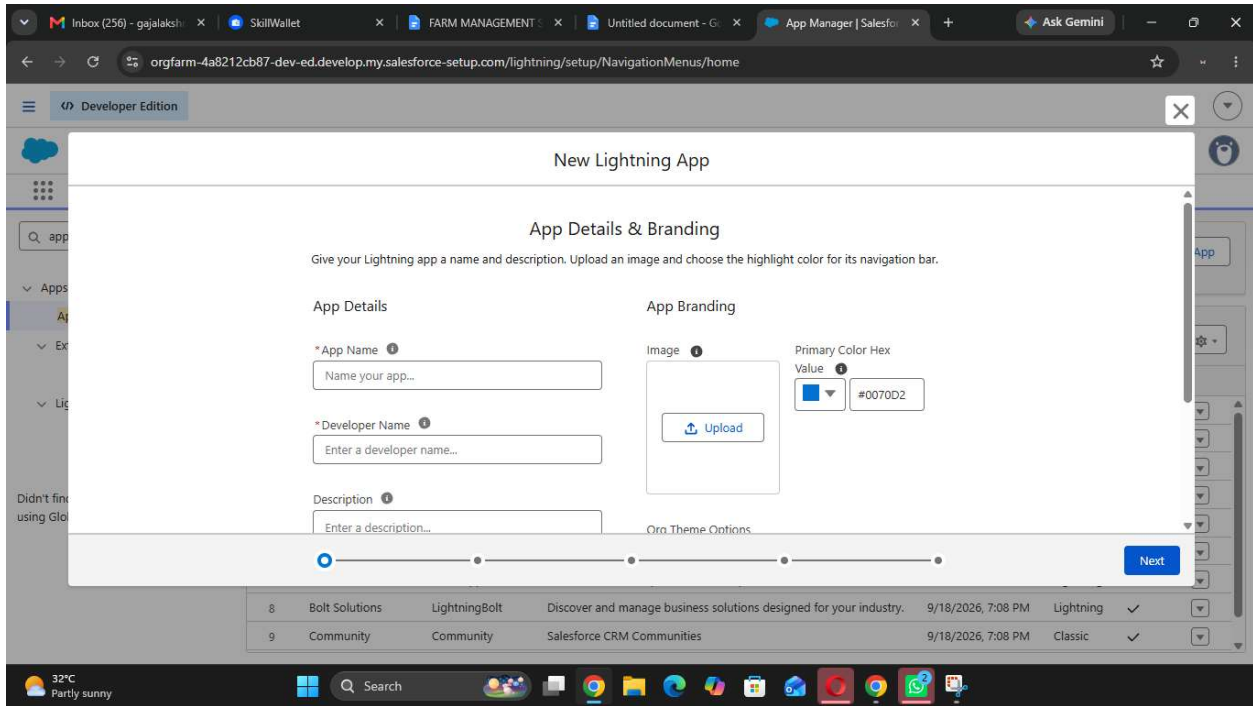
Apps in Salesforce are a group of tabs that help the application function by working together as a unit. It has a name, a logo, and a particular set of tabs. The simplest app usually has just two tabs.

Activity 1:

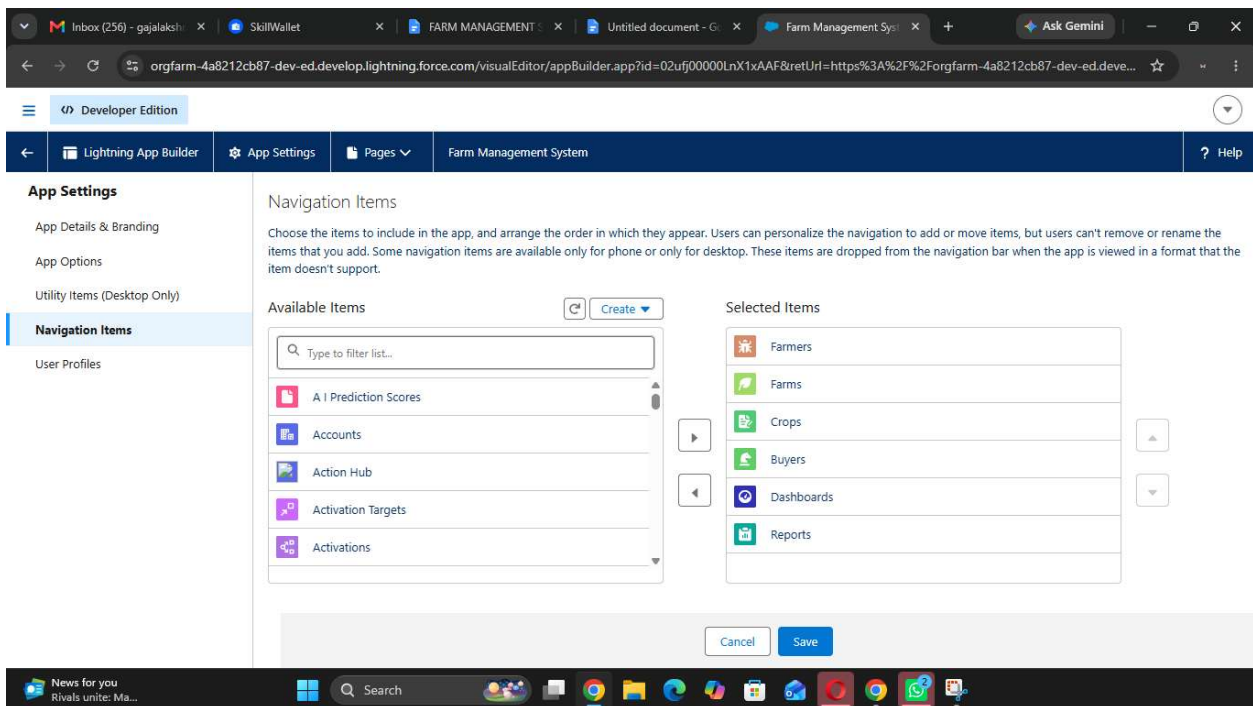
Create the Farm Management System app



- From Setup, enter App Manager in the Quick Find and select App Manager.
 - Click New Lightning App. Farm Management System as the App Name, then click Next
 - Under App Options, leave the default selections and click Next.
 - Under Utility Items, leave as is and click Next.
 - From Available Items, select Farm, Crop, Farmer, Buyer Reports, and Dashboards and move them to Selected Items. Click Next.
 - From Available Profiles, select System Administrator and move it to Selected Profiles. Click Save & Finish.
 - To verify your changes, click the App Launcher, type Farm Management System and select the Job Application app.
- Note:
1. App Launcher-Displays available apps.
 2. App Name-Displays the current selected app.



3. Navigation menu-Displays the tabs available inside the app.



Milestone 5 -

Fields and Relationship: What are fields?

Fields in Salesforce represent what the columns represent in relational databases. It can store data values which are required for a particular object in a record.

Activity 1:

Creation of fields for the Farm objects:

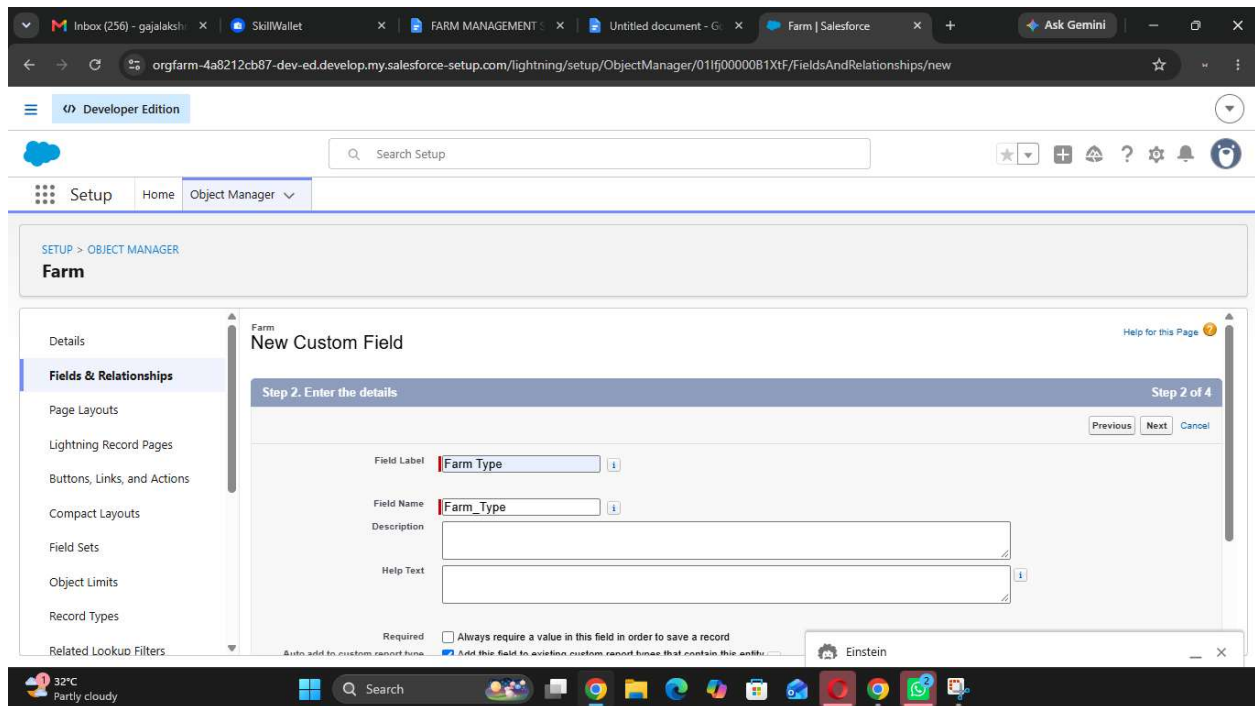
1. click the gear icon and select Setup. This launches Setup in a new tab.
2. Click the Object Manager tab next to Home.
3. Select Farm.
4. Select Fields & Relationships from the left navigation, and click New
5. Choose Data type as a picklist, Click next

The screenshot shows the Salesforce Setup interface. The left sidebar contains a navigation menu with options: Details, Fields & Relationships (selected), Page Layouts, Lightning Record Pages, Buttons, Links, and Actions, Compact Layouts, Field Sets, Object Limits, Record Types, and Related Lookups Filters. The main content area is titled 'Farm' and shows 'Fields & Relationships' with 16 items, sorted by Field Label. A table lists the fields:

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Actual Yield	Actual_Yield__c	Currency(18, 0)		
Created By	CreatedById	Lookup(User)		
Crop Name	Crop_Name__c	Text(255)		
Expected Yield per Acre	Expected_Yield_per_Acre__c	Formula (Currency)		
Farm	Farm__c	Lookup(Farm)		✓
Farm Name	Name	Text(80)		✓

6. Enter Field Label (Farm Type), and Enter Values as (Crop, Livestock, Mixed)

7. Next, Next Click Save.



Now let's create the other fields following Activity1 steps 1 to 4 and we must choose the data types of the fields carefully.

1. Next we Create another Picklist Field, Enter field label (Soil Quality) enter values (Organic, Cover Crops, Composting)
2. Again we Create Picklist field, Field Label (Irrigation type), Enter Values (Sprinkler irrigation, Drip irrigation, localized irrigation)
3. Choose data type as a text, enter field label (Location)
4. Choose data type as text, enter field label (size)
5. Create a Master Detail relationship with Farmer Object.

Activity 2:

Creation of fields for the Crop objects:

1. Choose Data type As a text, Enter Field Label name (Crop Name), Next save.
2. Select Data type as date, Enter Field label (Planting Date), then save.
3. Again we select data type as a date, Enter field label (Harvest date), save
4. Next We Create Lookup relationship on crop object related to Farm Object.
5. We select a data type as currency, Enter field Label (Actual Yield), Next Save it.
6. Select Data type as a Number, enter Field Label (Total Acres Harvested), Enter Maximum Length(20) and save.

7. Now We create Formula Field For That Select data Type As a Formula, Click Next.

The screenshot shows the Salesforce Setup interface in Developer Edition. The breadcrumb trail is **SETUP > OBJECT MANAGER**. The main heading is **Farm**. On the left sidebar, under **Fields & Relationships**, the following options are listed: Details, Fields & Relationships (selected), Page Layouts, Lightning Record Pages, Buttons, Links, and Actions, Compact Layouts, Field Sets, Object Limits, Record Types, and Related Lookup Filters. The main content area is titled **Data Type** and includes the instruction: "Specify the type of information that the custom field will contain." Below this, there are several radio button options: **None Selected**, **Auto Number**, **Formula** (selected), **Roll-Up Summary**, **Lookup Relationship**, **Master-Detail Relationship**, and **External Lookup Relationship**. Each option has a brief description of its functionality.

8. Enter Field Label (Expected Yield per Acre), Select formula return type as Currency, click Next.

9. Enter a Formula Actual Yield Divide By Total Acre Harvested, then click next and save.

The screenshot shows the same Salesforce Setup interface, but now the **Formula** data type is selected. The **Expected Yield per Acre (Currency)** label is entered. Below the label, the formula `Actual_Yield__c / Total_Acres_Harvested__c` is entered in the text area. The **Simple Formula** tab is active, and the **Insert Field** dropdown is set to **Farm**. The **Insert Operator** dropdown is set to **/**. The **Use Formula Assistant** button is visible. The bottom of the screen shows the Windows taskbar with the date and time as 32°C Clear.

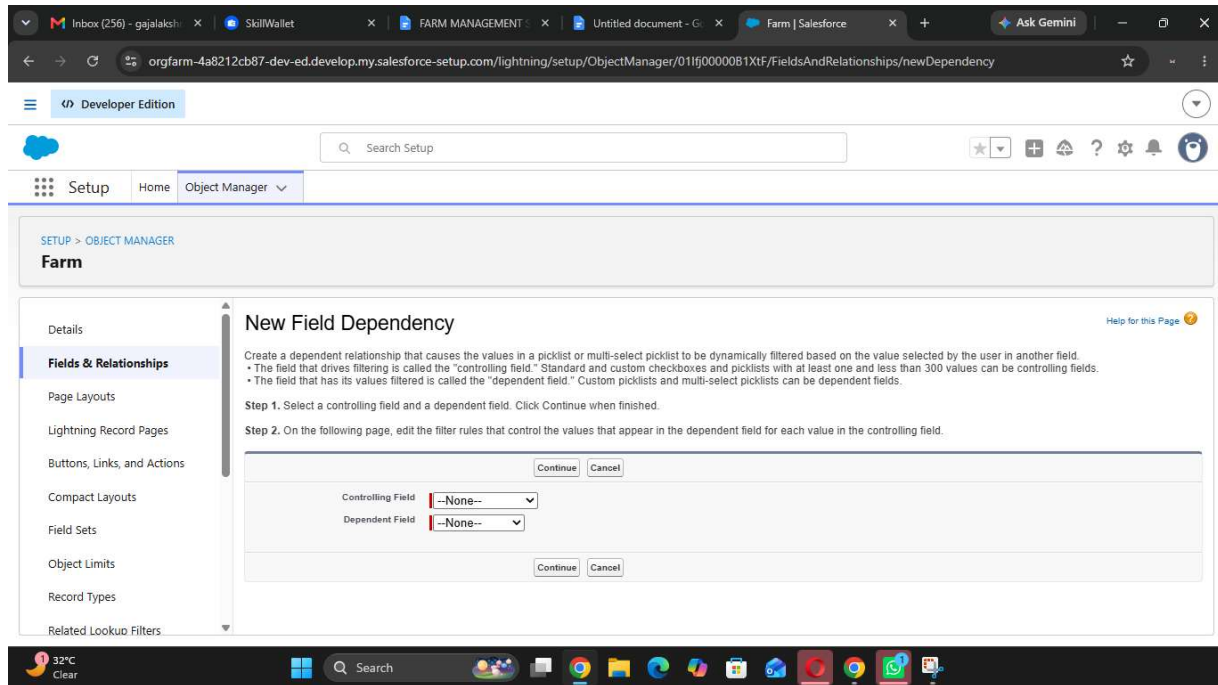
Activity 3:

Creation of fields for the Farmer objects:

1. Select Data type Phone, Enter field label (Contact Number), next Save.
2. Select Data type Picklist, Enter Field Label (State), Enter values (Telangana, Andhra Pradesh, Tamilnadu), save.
3. Choose Data type Picklist, Enter Field Label (District), Enter Values of district, next save.
4. Again select Data type as Picklist, Enter Field Label (Village), Enter Values of villages, next save.

Creation of field dependency :

1. Now create Field Dependency For District field, Open Object Manager, Choose Farmer object, select Fields & Relationships, On Right side We have Field Dependency option, select that option.
2. Click new, Select Controlling Field as State, Dependent Field As District, Click on Continue.
3. Select the appropriate Value in each column by double-clicking them.
4. For Ex. Telangana - Yadadri Bhongir.
5. Click Include Values. And it is also the same for Andhra Pradesh, Tamilnadu with its city.
6. Click Preview, then test the dependency by selecting a different State and viewing the associate Values available for Particular state.



7. Click Close to close the preview window, Click Save.

Activity 4:

Creation of fields for the Buyer objects:

1. Select Data type as a Phone, Enter a Field Label(Contact), Save.
2. Select Data type as an Email, Enter Field Label(Email), Next save.
3. Select Data type Text area, Enter Field Label(Address), Save it.
4. Select Data type Picklist, Field Label(Buyer Type), Enter values (Wholesale, Retail), Click Save.

Milestone 6- Profile:

What is a profile?

A profile is a group/collection of settings and permissions that define what a user can do in

salesforce. A profile controls “Object permissions, Field permissions, User permissions, Tab

settings, App settings, Apex class access, Visualforce page access, Page layouts, Record Types, Login hours & Login IP ranges.

A profile can be assigned to many users, but user can be assigned single profile at a time.

Activity 1:

Create a custom profile:

1. From setup , enter profiles in Quick Find box
2. Select profiles (Standard user).
3. Click clone.
4. For Profile, enter Sales Department.

Screenshot of the Salesforce Setup interface showing the Profiles page. The browser address bar shows the URL: `orgfarm-4a8212cb87-dev-ed.develop.my.salesforce-setup.com/lightning/setup/EnhancedProfiles/home`. The page title is "Profiles". The left sidebar shows the "Setup" menu with "Profiles" selected. The main content area displays a table of profiles:

Action	Profile Name	User License	Custom
Edit Delete	Sales Department	Salesforce	☒
Edit Delete	Salesforce API Only System Integrations	Salesforce Integration	☒
Edit Clone	Silver Partner User	Silver Partner	☐
Edit Clone	Solution Manager	Salesforce	☐
Edit Clone	Standard Platform User	Salesforce Platform	☐
Edit Clone	Standard User	Salesforce	☐
Edit Clone	System Administrator	Salesforce	☐

The table shows 1-7 of 7 profiles, with 0 selected. The "Sales Department" profile is highlighted.

Screenshot of the Salesforce Setup interface showing the Clone Profile page. The browser address bar shows the URL: `orgfarm-4a8212cb87-dev-ed.develop.my.salesforce-setup.com/lightning/setup/EnhancedProfiles/page?address=%2F_ui%2Fperms%2Fu%2Fprofile%2FProfileClone%2F%3F...`. The page title is "Clone Profile". The left sidebar shows the "Setup" menu with "Profiles" selected. The main content area displays a form for cloning a profile:

Enter the name of the new profile.

You must select an existing profile to clone from.

Existing Profile	User License	Profile Name
Standard User	Salesforce	Sales Department

Buttons: Save, Cancel

5. Click save.

Activity 2:

Create a custom profile-2:

Create a profile with the profile name as “ Farm Manager” .

1. From setup , enter profiles in Quick Find box
2. Select profiles (Standard user).
3. Click clone.

Milestone 7- Role

In Salesforce, roles are used to determine which users have access to certain data and functions within the system. They are also used to define the reporting hierarchy within an organization. Users with higher roles have greater access to data and more control over the system.

Activity-1

Creation of role

1. From the Quick find box search for the role and click on the roles option
2. select the set up roles option

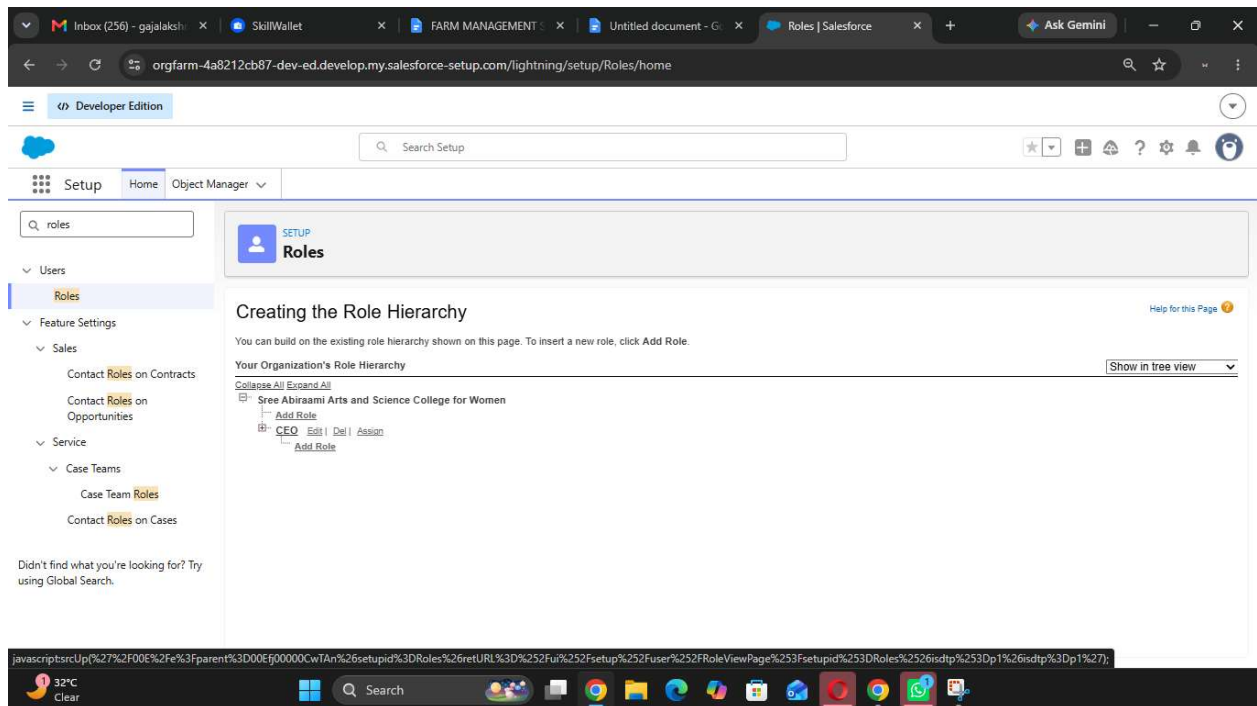
The screenshot displays the Salesforce Setup Roles page. The left sidebar shows the navigation menu with 'Roles' selected under 'Users'. The main content area is titled 'Understanding Roles' and includes a search bar for 'roles'. Below the search bar, there's a section for 'Sample Role Hierarchy' with a dropdown menu set to 'Territory-based Sample'. The hierarchy diagram shows the following structure:

- Executive Staff** (CEO, President, CFO, VP, Sales)
 - Western Sales Director** (Director of W. Sales)
 - Western Sales Rep** (CA Sales Rep, OR Sales Rep)
 - Eastern Sales Director** (Director of E. Sales)
 - Eastern Sales Rep** (NY Sales Rep, MA Sales Rep)
 - International Sales Director** (Director of Int'l Sales)
 - International Sales Rep** (Asian Sales Rep, European Sales Rep)

Each role box includes a brief description of its permissions, such as 'View & edit data, roll up forecasts, & generate reports for all users below' and 'Can't access data of other Executive Staff'.

At the bottom of the page, there are buttons for 'Set Up Roles' and 'Don't show this page again'.

3. Below the CEO click on add role and enter the label name as a " Farm Manager " and role name will be Automatically populated and click on save.



Milestone 8-User :

What is a user?

A user is anyone who logs into Salesforce. Users are employees at your company, such as sales reps, managers, and IT specialists, who need access to the company's records. Every user in

Salesforce has a user account. The user account identifies the user, and the user account settings determine what features and records the user can access.

NOTE- As Salesforce license can only be used by 2 Users at a time in Dev Org, so If you don't find

salesforce license then deactivate a user who has salesforce license Or change the license type from Salesforce to any other.

Activity 1:

To Create a user:

1. From Setup, enter Users in the Quick Find box, then select Users.
2. Click New User.

The screenshot shows the Salesforce Setup interface. The left sidebar contains a search bar with 'users' entered and a list of navigation items including 'Users', 'Analytics Groups', 'Permission Set Groups', 'Permission Sets', 'Profiles', 'Public Groups', 'Queues', 'Roles', and 'User Management Settings'. The main content area is titled 'Users' and includes a 'New User' button. Below this is a table of existing users.

Action	Full Name	Alias	Username	Role	Active	Profile
☐ Edit	Chatter Expert	Chatter	chatter.00f00000htrvaag.vdevvwtu0c7@chatter.salesforce.com		✓	Chatter Free User
☐ Edit	Department Sales	adeoa	divvasiva07@gmail.com	Farm Manager	✓	Farm Manager
☐ Edit	EPIC_OrgFarm	OEPIC	epic.127ae741a508@orgfarm.salesforce.com		✓	System Administrator
☐ Edit	Manager_Farm	fman	divya07@gmail.com	Farm Manager	✓	Farm Manager
☐ Edit	B. Gajalakshmi	gali	gajalakshmi031209.49493o131a43@agentforce.com		✓	System Administrator
☐ Edit	User_Integration	inteq	integration@00f00000htrvaag.com		✓	Analytics Cloud Integration User
☐ Edit	User_Security	seo	insightsecurity@00f00000htrvaag.com		✓	Analytics Cloud Security User

3. Enter First name as Farm and last name as Manager.
4. Enter the user's name and email address and a unique username in the form of an email address. By default, the username is the same as the email address.
5. Select the role Farm Manager.
6. Select user License as Standard Platform User.
7. Select profile(Farm Manager).
8. Click save.

Activity 2:

Create a user with a username as “Sales Department”, and assign him the sales Manager profile. Follow the steps from above Activity

Milestone 9-Sharing Rules:

What are Sharing Rules?

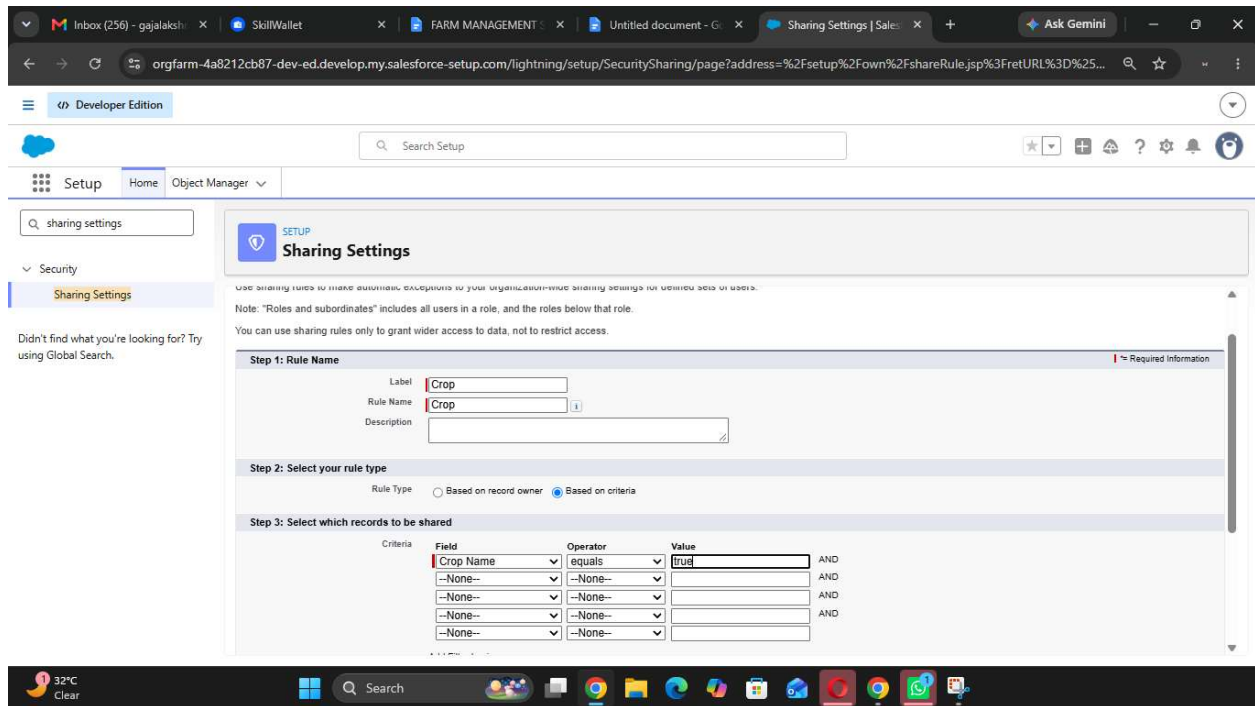
Sharing rules help users to share records based on conditions. It is basically created for objects whose organization-wide defaults (OWD) are set to public read-only or private because sharing rules can only extend the access and not restrict it.

Types of sharing rules,

•Owner-based Sharing Rules •Criteria-based Sharing Rules

Activity 1:**Create a sharing rule**

- 1) Goto Sharing Settings, which can be found under the Quick Find section.
- 2) Scroll down and find the candidate object where a sharing rule needs to be added, and then click on New to create a new sharing rule.
- 3) Add the label of the sharing rule you want to make.
- 4) Select your rule type based on the criteria.
- 5) Select the field can join immediately check field from the Crop object.
- 6) Select the Crop as equal and value is True.
- 7) And in selecting the users to share with the section select roles and in that select Farm Manager.
- 8) And in the section of select the level of access for the users give the access Read/Write.
- 9) And save the rule.



Activity 2:

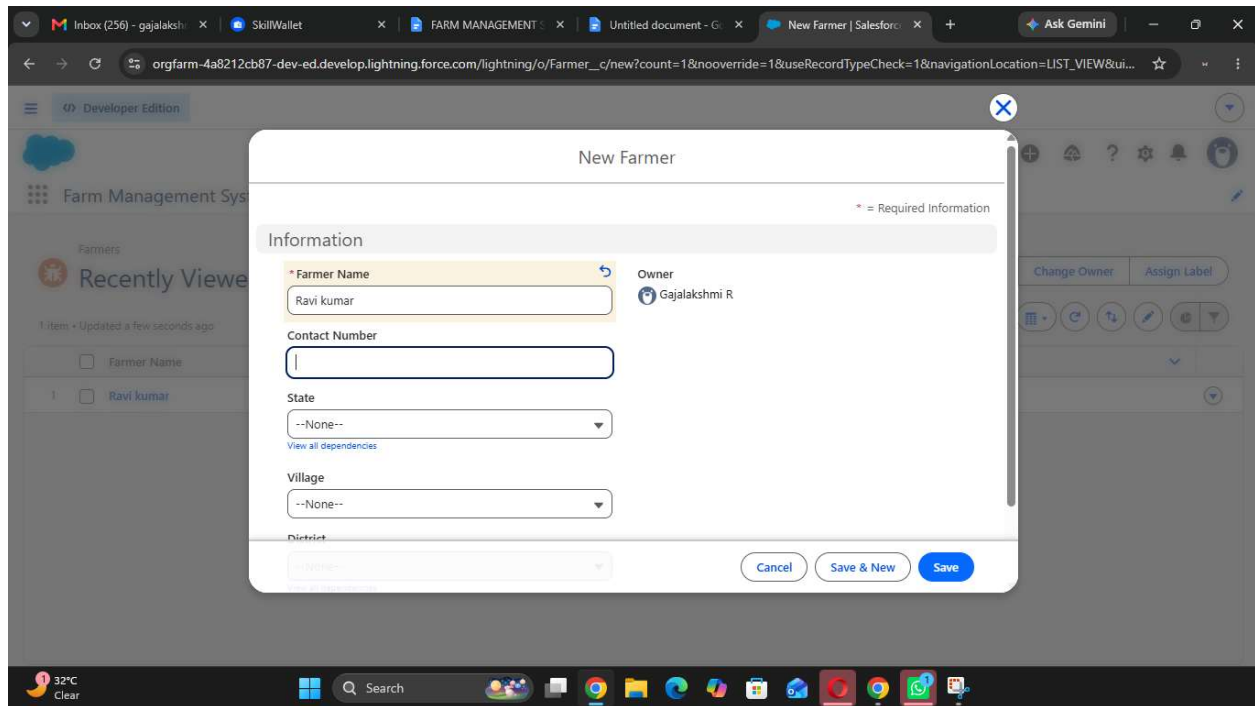
Create a Sharing Rule to Share the records of Buyer to Sales department with the Access of Read/Write.

Follow the steps from above Activity.

Milestone-10: User Adoption

Activity 1: Create a Record(Farmer)

- Click on App Launcher on left side of screen.
- Search Farm Management & click on it.
- Click on Farmer Tab.
- Click new and fill details & Save.



Milestone 11-Reports:

What are Reports?

A report is a list of records that meet the criteria you define. It's displayed in rows and columns, and can be filtered, grouped, or displayed in a graphical chart. Every report is stored in a folder. Folders can be public, hidden, or shared, and can be set to read-only or read/write.

Activity 1:

Create a report:

Create a report that displays rating of the account and which has type and account name.

- 1) Click on app launcher search for reports.
- 2) Click on the new report and select the category has Farmer with Farms.

Activity 2:

Create a Report using the Objects Crops, Buyers. Follow the steps from above Activity.

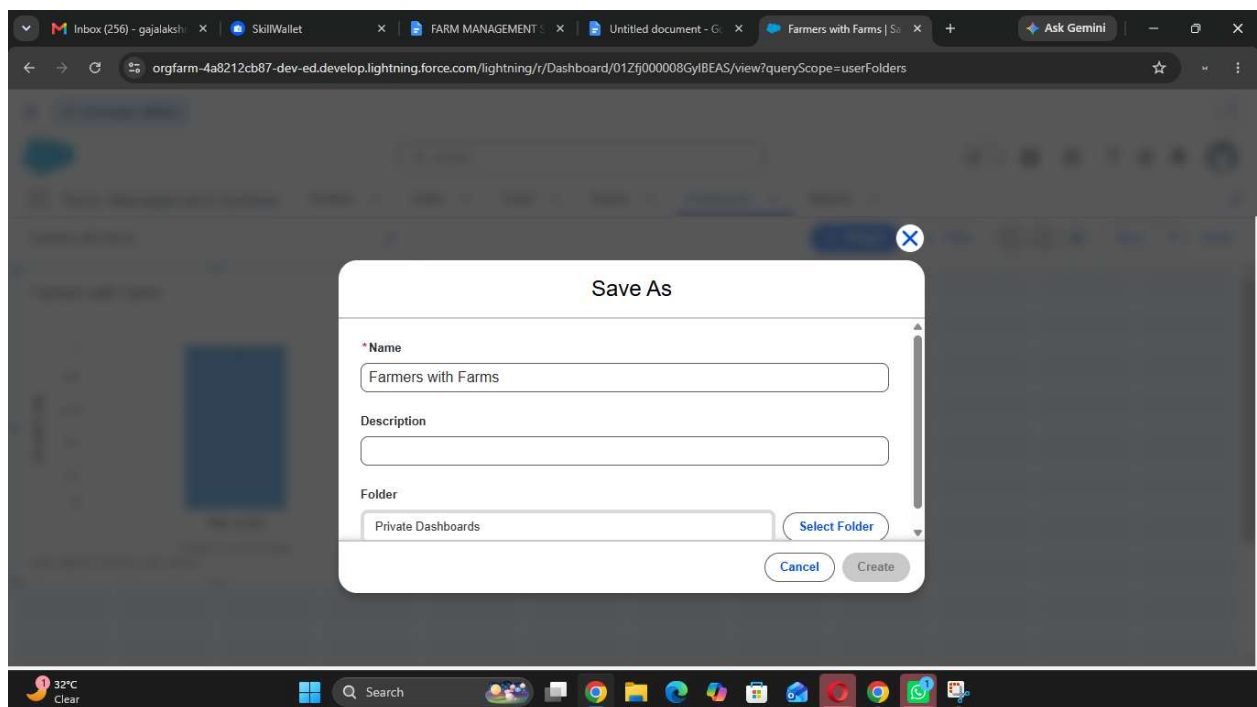
Milestone 12-Dashboard:

Dashboards provide more insights than reports as they combine the data from many reports and show a summarized result. Looking at many reports at a time gives the flexibility of combining the results from them quickly. Also summaries in dashboards help us decide on action plans quicker. The dashboards can contain charts, graphs and Tabular data.

Activity 1:

Create a Dashboard:

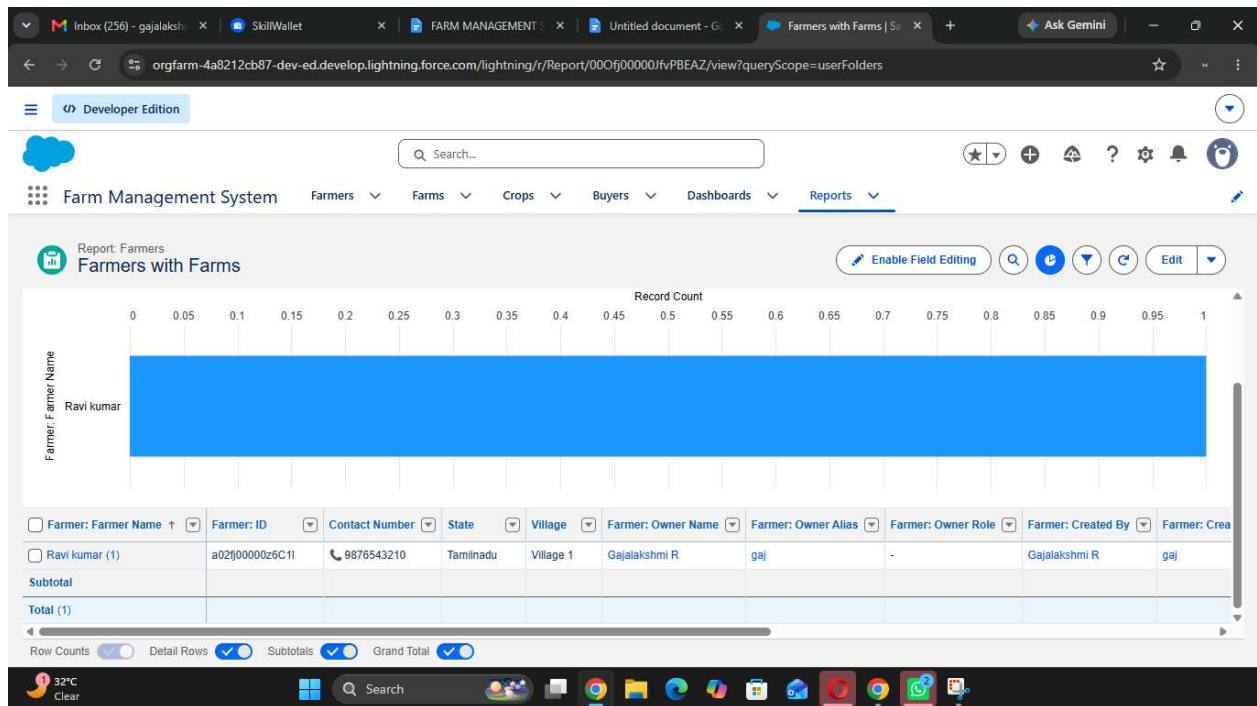
1. Click the Dashboard tab.
2. Click New Dashboard.
3. Name the dashboard Farmers with farms and click Create.
4. Click +Component.
5. Select the Farmers with Farms and click Select.
6. Select the Gauge chart and click Add.
7. Click Save and then Done.



Milestone-13: View Report and Dashboard:

Activity 1: Report

- Click on App Launcher on left side of screen.
- Search Farm Management System & click on it.
- Click on Reports Tab.
- Click on Farmers with Farms & see records



Activity 2: Dashboard

- Click on App Launcher on left side of screen.
- Search Farm Management system & click on it.
- Click on Dashboard Tab.
- Click Farmers With farms & see records.

